

Health Inequality and Public Policy

(2)

What is health inequality?

Monday, 22. September 2025

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Health Communication

Your experiences matter!

In my bachelor's thesis at the University of Lucerne, I am investigating which **communication problems occur between patients with a migration background, and medical professionals** - and how they can be improved.

Who am i looking for?

Students with a migration background who have had experiences with the Swiss healthcare system (e.g. doctors visits, hospital visits, etc.)

What can you expect?

a short interview (30 - 45 min)
online or in person



Interested or have
questions? Scan the QR
code and contact me.



TWO KEY DEFINITIONS

Health inequities

The unfair and avoidable or remediable systematic differences in health among population groups defined socially, economically, demographically or geographically. Health inequities arise fundamentally from the unequal allocation of power, money and resources, which manifest in the unequal distribution of social determinants.

Health inequalities (USA: disparities)

The observable differences in health between population groups. Health inequality is the metric by which health inequity can be assessed, meaning characteristics that stratify people's power and access to opportunities should most closely reflect likely areas of unfair health differences.

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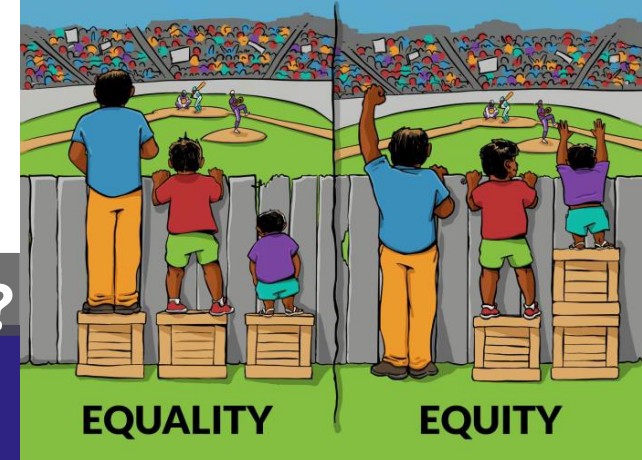
Accurate?

Health inequities

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Which definition do you find more convincing, and why?

DEFINITION OF HEALTH EQUITY IN SWITZERLAND

Swiss Federal Office of Public Health (FOPH):

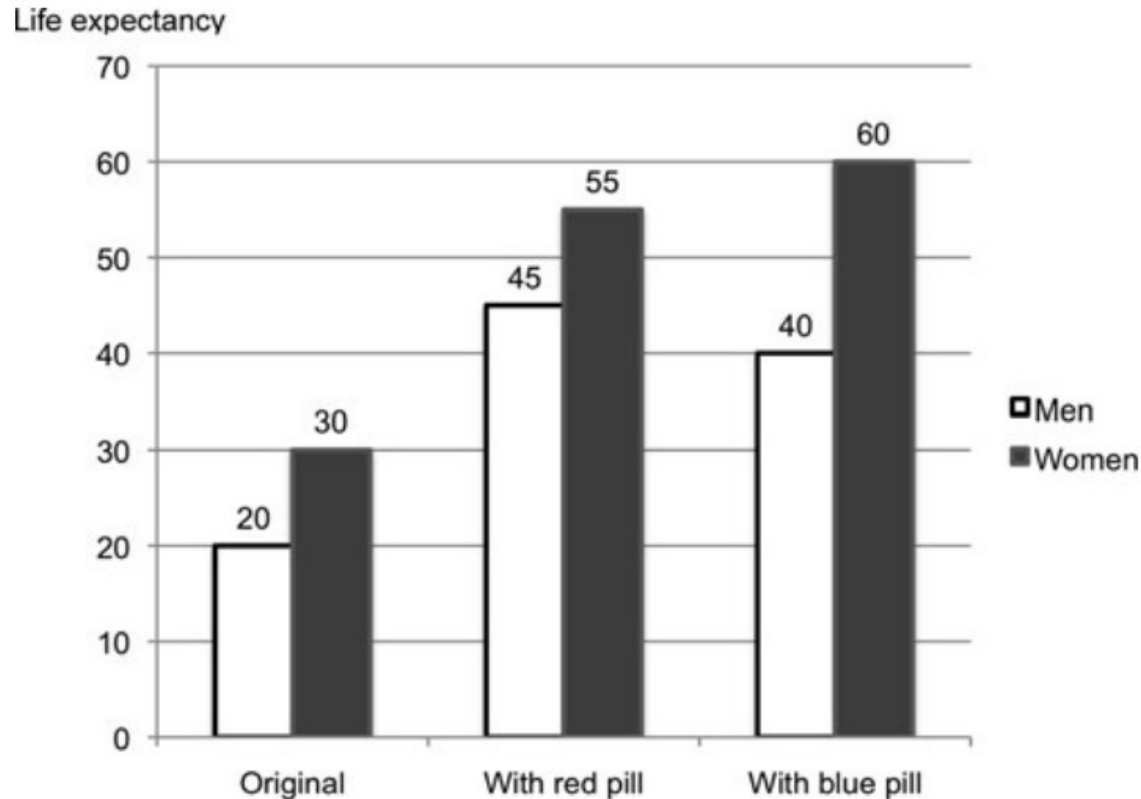
“Ensuring **health equity** means ensuring that everyone has equal opportunities to develop, maintain and if necessary restore their health.”

(«**Gesundheitliche Chancengleichheit** bedeutet, dass alle die gleichen Möglichkeiten zur Entwicklung, Erhaltung und falls nötig Wiederherstellung ihrer Gesundheit haben.»)

“The promotion of health equity and equitable access to healthcare are guiding principles of Switzerland’s health and integration policy. Through its activities, the FOPH also aims to enhance quality and efficiency in the healthcare system.”

Contrast the FOPH’s definition with the WHO definitions on health inequities and health inequalities:
Similarities?

MEASURING HEALTH INEQUALITY – ASADA’S EXPERIMENT



Scenario: Imagine two equal groups (e.g. men and women). Their life expectancy starts at 20 and 30. Everyone then takes one of two pills:

- The **red pill** adds the same number of years to both groups.
- The **blue pill** multiplies life expectancy by the same proportion for both groups.

Both pills raise *average* life expectancy by the same amount (+25).

But how does health inequality change...
...with the **red pill**?
...with the **blue pill**?

ABSOLUTE AND RELATIVE “GAP” MEASURES

Rate Difference (RD): $y1 - y2$

where $y1$ is the health status of the least healthy group and $y2$ is the health status of the most healthy.

More complex version (regression-based): Slope Index of Inequality

Rate Ratio (RR): $y1 / y2$

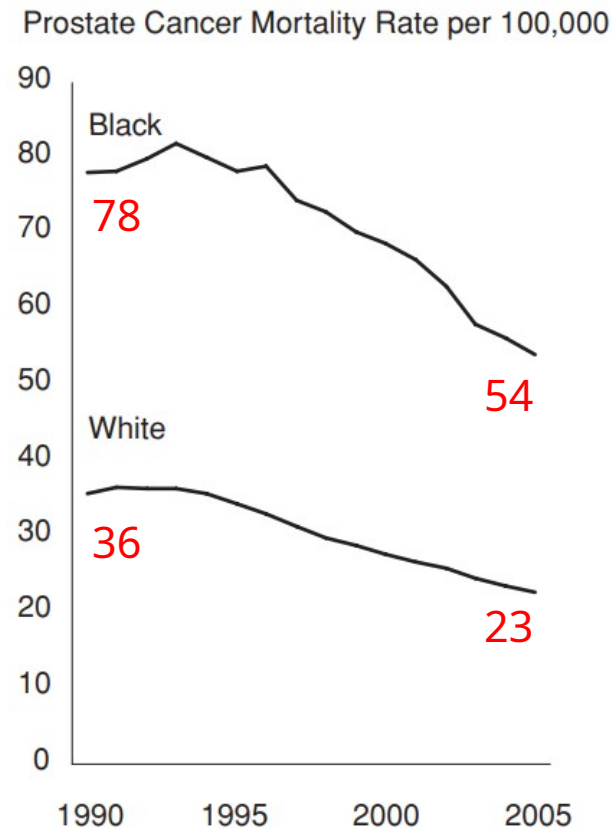
where $y1$ is the health status of the least healthy group and $y2$ is the health status of the most healthy.

More complex version (regression-based): Relative Index of Inequality

EXERCISE: ABSOLUTE AND RELATIVE HEALTH INEQUALITIES

Trends in Age-Adjusted Prostate Cancer Mortality among Black and White Males in the US, 1990-2005

Source: Harper et al. (2010)

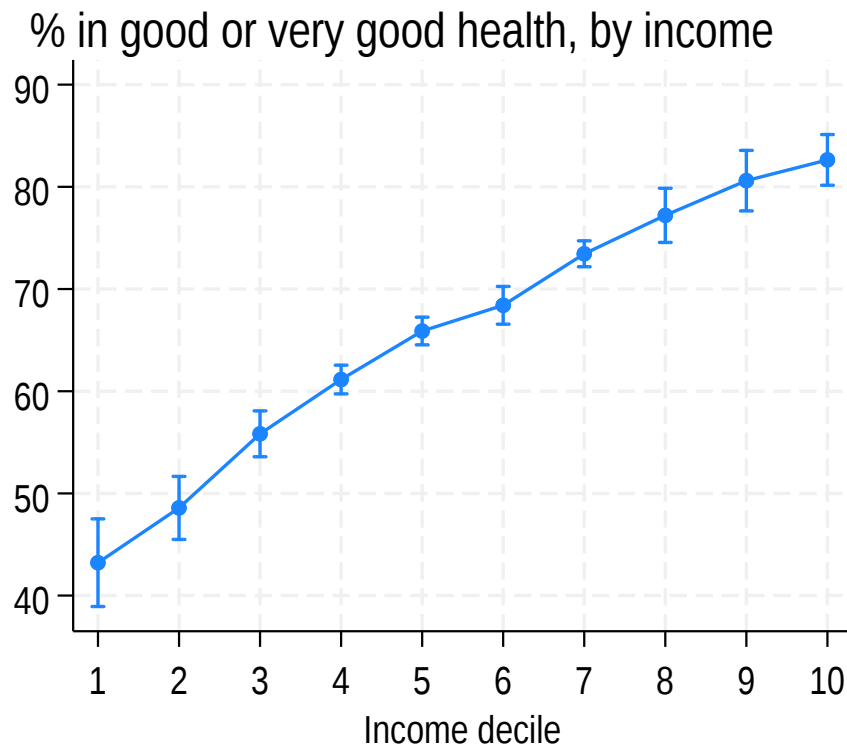


(1) Calculate the rate difference and the rate ratio in prostate cancer mortality between Black and White men in 1990 and 2005.

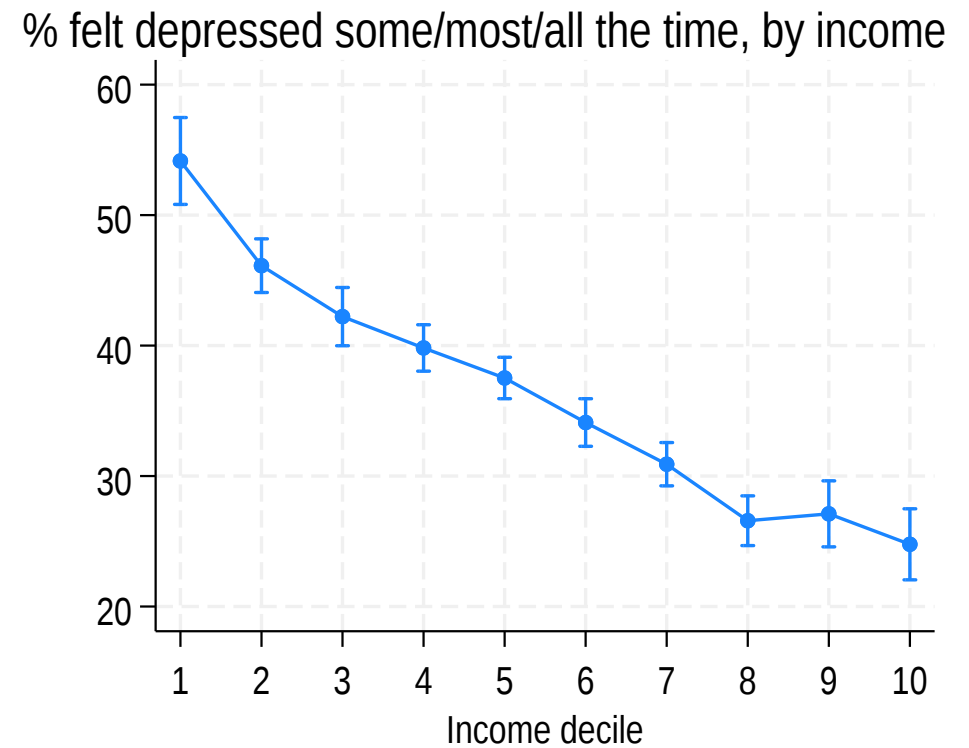
(2) How do you evaluate the observed trend in health inequality between 1990 and 2005?

HEALTH GAPS VS. GRADIENTS

“The social gradient in health: This means people and communities have progressively better health, the higher their socioeconomic position or conditions”
(WHO 2025)



Source: ESS round 11 (2023-24), 28 countries, n=34,848, with 95%-CI



Source: ESS round 11 (2023-24), 28 countries, n=34,784, with 95%-CI

IMPLICATIONS FOR WRITING ABOUT HEALTH INEQUALITY

Review | [Open access](#) | Published: 02 September 2025

Writing about health inequality: recommendations for accurate and impactful presentation of evidence

[Nicole Bergen](#), [Katherine Kirkby](#), [Aluisio J D Barros](#), [Paula Braveman](#), [Peter Goldblatt](#), [Theadora Swift Koller](#),
[Oscar J Mujica](#), [Devaki Nambiar](#), [Owen O'Donnell](#), [Anne Schlottheuber](#), [Vivian Welch](#) & [Ahmad Reza Hosseinpoor](#) ✉

International Journal for Equity in Health **24**, Article number: 231 (2025) | [Cite this article](#)

Recommendation 1: Use terminology deliberately and consistently

- Health inequality vs health inequity; dimensions of inequality; sex/gender; race/ethnicity

Recommendation 2: Present statistical content accurately and with sufficient detail

- Units of measurement (absolute vs relative; percentage vs percentage point); association vs causation

Recommendation 3: Adhere to reporting guidelines and best practices

- Report both absolute and relative measures, average values, (sub)groups size, etc.

Recommendation 4: Respect and uphold the interests of affected communities through inclusive writing practices

- Affirming / strength-based language, co-writing with communities represented in research where possible

DATA SOURCES - INTERNATIONAL



OECD > Topics > Health inequalities

Health inequalities

Large inequalities in health and life expectancy exist between different groups of people. These health inequalities can be related to the unequal exposure to health risk factors, such as smoking, as well as unequal access to healthcare, among other aspects. They are often linked to a range of social, economic and environmental factors which require large-scale government action to

eurostat

Detailed datasets

- Health (hlth)
 - Health status (hlth_state)
 - Health determinants (hlth_det) 
 - Health care (hlth_care)
 - Causes of death (hlth_cdeath) 
 - Health and safety at work (hsw)



World Health Organization

Health Topics ▾ Countries ▾ Newsroom ▾ Emergencies ▾ Data ▾ About WHO ▾

Health Inequality Monitor

Explore health inequality monitoring evidence, tools, resources and training

Home / Data / Health Inequality Monitor / Data

Health Inequality Data Repository

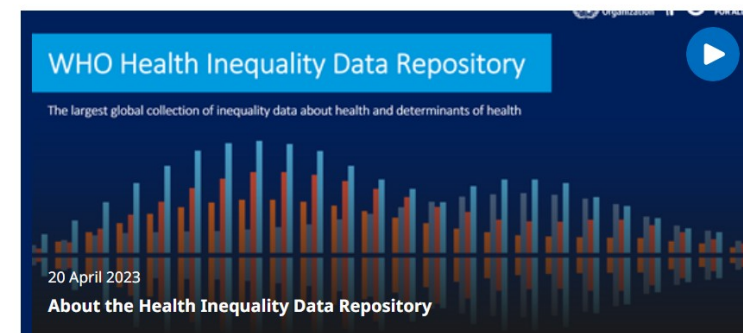
Disaggregated data – which show how health or other aspects of life are experienced by people of different ages, economic status, education levels, place of residence, sex, and other characteristics – are a vital part of advancing equity.

Disaggregated data can serve as an evidence base for equity-oriented policies, programmes and practices that aim to close existing gaps and achieve good health and well-being for all.

The Health Inequality Data Repository contains datasets of disaggregated data covering diverse topics and dimensions of inequality, from a variety of publicly available data sources. It aims to make disaggregated data more accessible and navigable to diverse global audiences.

Please note that some datasets are from external published sources and are

<https://www.oecd.org/en/topics/sub-issues/health-inequalities.html>
<https://ec.europa.eu/eurostat/web/health/database>
<https://www.who.int/data/inequality-monitor/data>



WHO Health Inequality Data Repository

The largest global collection of inequality data about health and determinants of health

20 April 2023

About the Health Inequality Data Repository

DATA SOURCES - SWITZERLAND

Indicators on Healthcare Equity in Switzerland New Evidence and Challenges

Final report on behalf of the Federal Office of Public Health 2021

Jacques Spycher, Kevin Morisod, Yves Eggli, Karine Moschetti, Marie-Annick Le Pogam,
Isabelle Peytremann-Bridevaux, Patrick Bodenmann, Joachim Marti

bag.admin.ch/dam/de/sd-web/ejzz0Ebd9ndr/report-indicators-on-healthcare-equity.pdf

<https://www.obsan.admin.ch/de/publikationen/2023/dheitliche-chancengleichheit-datenquellen>



Schweizerisches Gesundheitsobservatorium
Observatoire suisse de la santé
Osservatorio svizzero della salute
Swiss Health Observatory

Gesundheitliche Chancengleichheit: Datenquellen

Bericht im Auftrag des Bundesamts für Gesundheit (BAG)

APPLICATION – RECAP WEEK 1

Milestone week 1: Getting started, loading the ISSP/ESS datasets

- R/Stata code on Moodle
- Questions, tips, difficulties?

APPLICATION WEEK 2 – FIRST DESCRIPTIVE INDICATORS

Milestone today: First basic descriptive health inequality indicators in the ISSP/ESS

- Task 1: Find/choose a suitable **outcome measure** (e.g. subjective health status, depression, etc.)
- Task 2: Create a **health gap** measure
 - Gap between two groups: Rate difference and rate ratio
 - Use a categorical group measure: men vs women, high vs low education, urban vs rural, migrant vs non-migrant background, etc.
- Task 3: Create a plot that shows a **health gradient**
 - Use a continuous/ordinal measure: for example, income, financial difficulties in family when growing up, subjective social status, etc.

OUTLOOK NEXT WEEK

3 29.09.2025 Explanations of health inequality

Reading: Bartley, M., & Kelly-Irving, M. (2025). *Health Inequality. An Introduction to Concepts, Theories and Methods* (3rd ed.). Polity Press.

Chapter 4 “Explanatory Models I: Behavioural Explanations”, pp. 75-93,

Chapter 5 “Explanatory Models II: Material Explanations”, pp. 94-110,

Chapter 6 “Explanatory Models III: Psychosocioal Factors”, pp. 111-126.

Applied focus: Finding items to analyse behavioural, material, psychosocial explanations in the ISSP/ESS

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